UNIVERSITY of York
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Computer Science Department

THE LIBRARY CASE STUDY

System Description

You have been contracted to develop a computer system for a university library.
The library offers a collection of titles, which includes books and journals. For each title offered, a unique id and its title are recorded. If the title is a book, the author is also recorded, whereas if the title is a journal the editor is also recorded.

The library may have several copies of a given book, each with a unique copy id but only one copy of each journal. Only book copies are borrowable, whereas journals can never leave the premises of the library. A library member may borrow up to 6 borrowable items at a time, for a period of three weeks each.

When a library member requests to borrow a particular book copy, the system checks that the potential borrower is a member of the library, and that they do not already have the maximum permitted number of borrowable items on loan. If both checks succeed, the system records that this library member has this copy of the book on loan. Otherwise it refuses the loan.

Summary and thoughts

A library needs to maintain electronic records of:
- The titles collection that it offers;
- The particular items that can be borrowed from the library (borrowable items);
- Its members.

The library's titles collection includes books and journals. Borrowable items are only the book copies, i.e. copies of book titles. There might several copies of each book however journals only come in a single copy and are not borrowable.

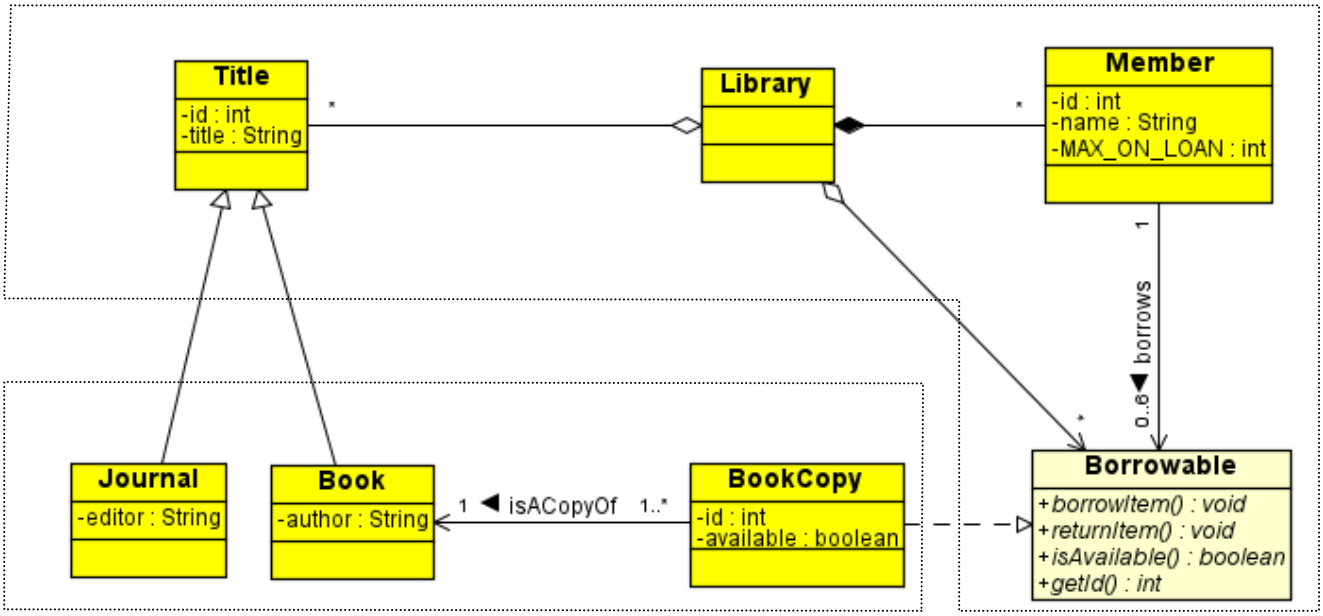
Main Classes identified together with their attributes:

Title	**Journal**	**Book**	**Member**	**BookCopy**
▪ id ▪ title	<u>is a kind of</u> title Additional attribute: ▪ editor	<u>is a kind of</u> title Additional attribute: ▪ author	▪ id ▪ name ▪ MAX_ON_LOAN	is a copy of a particular Book Attributes: ▪ id ▪ available

Knowing that an item is borrowable gives us information about the behaviour of this item:
- It can be borrowed;
- It can be returned;
- We should be able to know at any point whether it is available or not.

To be able to enforce these three desired behaviours on any kind of borrowable item we need a **Borrowable** interface with the following methods:
- void borrowItem()
- void returnItem()
- boolean isAvailable()

Based on the Class Diagram provided below, carry out all the numbered exercises.
Consult your notes whenever necessary.



1. Write the code for the `Title` and `Member` classes. Save them as `Title.java` and `Member.java`, respectively. Compile.
(There is no point in executing the above because there is no `main` method!)

2. Write the code for the `Journal` and `Book` classes. Compile.
How is the **inheritance** with `Title` implemented?
Are there cases of **polymorphism:** inherited methods that need to be overridden/extended?

3. Write the code for the `Borrowable` **interface**. Save it as `Borrowable.java`. Compile.

4. Write the code for the `BookCopy` class. Save it as `BookCopy.java`. Compile.
How is the **association** with `Book` implemented?
How is the **realisation** of the `Borrowable` interface implemented?

5. Complete the `Member` class. Compile.
How is the **association** with `Borrowable` implemented?

6. Write the code for the `Library` class. Make sure that you:

- **a.** Define a List or Map attribute named `members` that will hold the member objects representing the members of the library.
- **b.** Define a List or Map attribute named `titles` that will hold the title objects that the library offers.
- **c.** Define a List or Map attribute named `borrowables` that will hold the book copy objects (or any other objects in the future) that can be *borrowed* from the library.
- **d.** Develop the default constructor of the `Library` class. Use the `ArrayList` as a realisation of the `List` interface or the `HashMap` as a realisation of the `Map` interface.
- **e.** Develop the method `public void addMember(int id, String name)` that will add a member to the library (i.e. to `members`).
  How is the **composition** between `Library` and `Member` implemented?
- **f.** Develop the method `public void addTitle(Title t)` that will add a title to the collection of the library (i.e. to `titles`).
- **g.** Develop the method `public void addBorrowable(Borrowable b)` that will add a borrowable item to the collection of the library (i.e. to `borrowables`).
- **h.** Develop the method `public void displayAllMembers()`, that will display all library members.
- **i.** Develop the method `public void displayAllBorrowables()`, that will display all borrowable library items.
- **j.** Develop the method `public void displayBorrowedItems()`, that will display all library items that are currently borrowed, i.e. not available.

7. Create a `LibraryProgram` class that will contain the `main` method, in which you will instantiate a library and "play around" with it:

- **a.** Create a new library and name it `lib`.
- **b.** Add 3 titles to the library: 2 books and 1 journal.
- **c.** Add 3 book copies to the library, 2 of them corresponding to the same title.
- **d.** Add 2 members to the library.
- **e.** Have one of the members borrow 2 book copies.
- **f.** Use the display methods of the library to display all relevant information.
- **g.** Have the member return one of the book copies and display all info again.

8. As a last step extend your implementation to include `Exceptions` whenever appropriate. Issues for discussion:

- **a.** Creating your own Exception subclasses
- **b.** Throwing Exceptions
- **c.** Handling Exceptions

Issues for Discussion:

- How would the design and implementation be extended to include DVDs that can be borrowed? How would the design and implementation be extended to include different types of members (e.g. staff members)?
  ⇒ The importance of an extensible design!
- Reasons for implementing the `toString()` method.

Additional requirements for further practice

The system should, of course, keep track of when book copies are borrowed and returned. It should also allow any user to (a) browse and search through all offered titles (based on author, title, etc.), and (b) check whether a book copy is available for loan.

Two new requirements (not supported by the current system) state that library members can (a) reserve an unavailable book copy, so that they borrow it upon its return, and (b) extend the loan period of a book copy they already have by one week, if the book copy is not reserved to be borrowed by another member.

Finally, at any point when necessary, the librarian may update the titles' catalogue.